INCH-POUND

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22 August 2000
SUPERSEDING
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DETAIL SPECIFICATION

# CONNECTORS, ELECTRICAL, MINIATURE, COAXIAL, ENVIRONMENT RESISTANT TYPE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

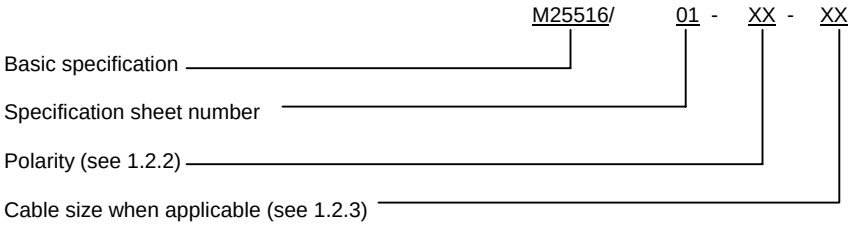
## 1. SCOPE

1.1 Scope. This specification covers a series of miniature, moisture proof connectors suitable for operation with shielded and unshielded cable up to 750 volts rms under severe environmental conditions.

1.2 Classification. Connectors covered by this specification should be of the following series, classes, and types, as specified (see 3.1).

| | | |
|---|---|---|
| Series I | - | Plugs and receptacles, cable termination. |
| Series II | - | Plugs and receptacles, permanent crimp, (using a standard MIL-C-22520 tool). |
| Series III | - | Adapters, all types. |
| Series IV | - | Plugs and receptacles, solder termination. |
| Class G | - | General purpose. |
| Class H | - | Hermetic, sealed cable termination. |
| Class Y | - | Hermetic, adapters. |
| Type 1 | - | Shield termination, continuous circuit, grounded. |
| Type 2 | - | Shield termination, continuous circuit, isolated ground. |
| Type 3 | - | Shield termination, interrupted ground. |
| Type 4 | - | No shield termination. |
| Type 5 | - | Class Y, grounded outer circuit. |
| Type 6 | - | Class Y, ungrounded outer circuit. |

1.2.1 Part or Identifying Number (PIN). The military PIN should consist of the letter “M” followed by the basic specification sheet number and a sequentially assigned dash number which is used to designate the polarity and a letter to designate the cable size as shown in the following example:

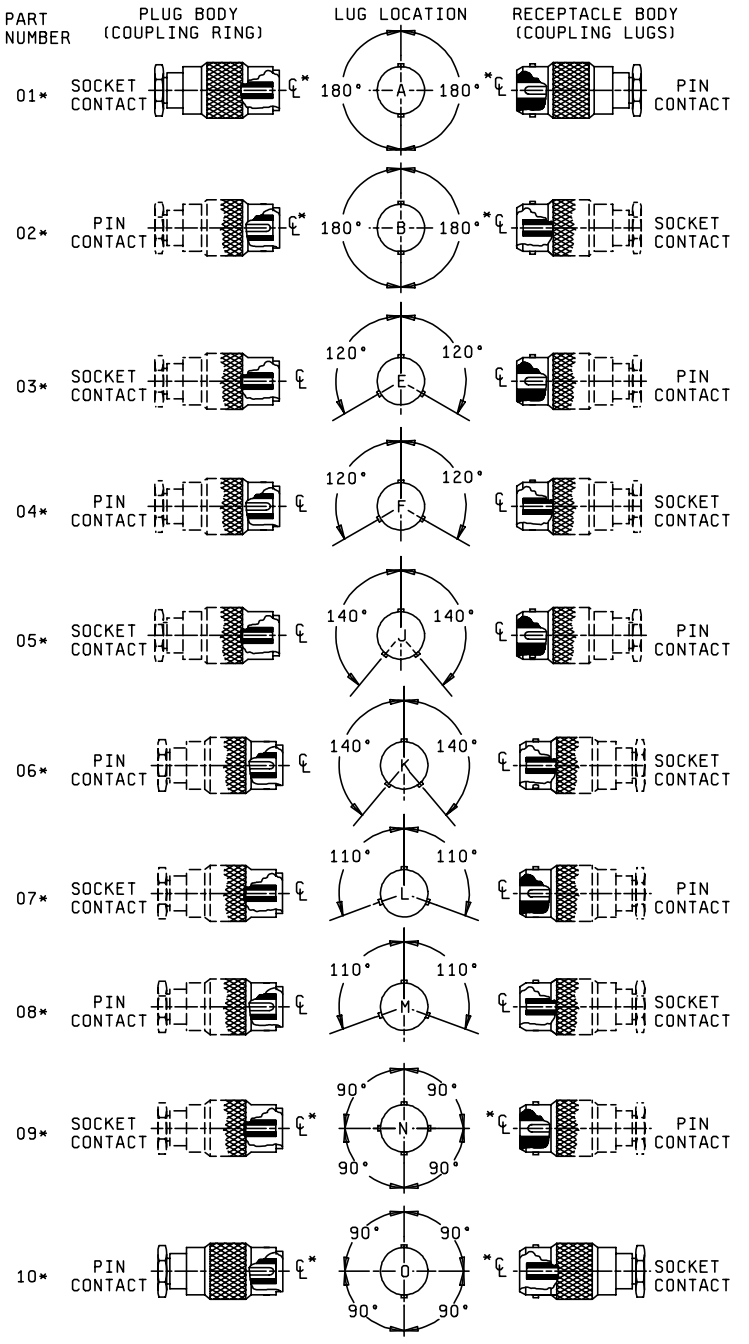



1.2.2 Polarity. The mating interface of connectors covered by this specification should be polarized as specified on figure 1 (see 3.1). Where connector configuration includes minor variables such as mounting features, solder terminations, etc. The first digit in the polarity dash number may be changed to show the variation. Dash number for polarities should be as specified in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, Attn: VAI, 3990 East Broad Street, Columbus, Ohio, 43216-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Slot in coupling rings mate with receptacle lugs)

Receptacles (not shown) have coupling lugs

* Cancelled without replacement.

FIGURE 1. Polarity chart.



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TABLE I. Polarities.

| Dash number | Polarity | Uses |
|---|---|---|
| 03 | E | Basic configuration of each slash sheet |
| 04 | F | |
| 05 | J | |
| 06 | K | |
| 07 | L | |
| 08 | M | |
| 13 | E | Use for configuration variations |
| 14 | F | |
| etc. | etc. | |

1.2.3 Cable size. Connectors covered by this specification should be capable of terminating cables as designated per Appendix 1 when applicable (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

O-F-499 - Flux, Brazing, Silver Alloy, Low Melting Point.
QQ-S-365 - Silver Plating, Electrodeposited: General Requirements For.

DEPARTMENT OF DEFENSE

MIL-DTL-5606 - Hydraulic Fluid, Petroleum Base; Aircraft: Missile and Ordnance.
MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-PRF-6081 - Lubricating Oil, Jet Engine.
MIL-PRF-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-22520 - Crimping Tools, Terminal, Hand or Power Actuated, Wire Termination, and Tool Kits.
MIL-C-26074 - Coating, Electroless Nickel, Requirements For.

(See Supplement 1 for applicable specification sheets.)



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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

DEPARTMENT OF DEFENSE

MIL-STD-100 - Engineering Drawing.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-1344 - Test Methods for Electrical Connectors.

HANDBOOK

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - General Guidelines for Electronic Equipment.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents, which are DoD adopted, are those listed in the issue DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI J-STD-004 - Requirements for Soldering Fluxes.
ANSI J-STD-005 - Requirements for Soldering Pastes.
ANSI J-STD-006 - Requirements for Electronic Grade Solder Alloys, and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108 - Standard and Specification for Steel Bars, Carbon, Cold-Finished Standard Quality.
ASTM A484 - Steel, Bars, Billets and Forgings, Stainless.
ASTM B152 - Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar.
ASTM B488 - Electrodeposited Coating of Gold for Engineering Uses.
ASTM B36 - Brass Plate, Sheet, Strip and Rolled Bar.
ASTM B121 - Lead Brass Plate, Sheet, Strip and Rolled Bar.
ASTM B124 - Copper and Copper Alloy Forging Rod, Bar and Shapes.
ASTM B194 - Copper Beryllium Plate, Sheet, Strip and Rolled Bar.
ASTM B196 - Copper Beryllium Alloy Rod and Bar.
ASTM B197 - Copper Beryllium Alloy Wire.
ASTM D471 - Standard Test Method for Rubber Property-Effect of Liquids.
ASTM D910 - Standard Specification for Aviation Gasolines.

(Applications for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS31971 - Pin, Gage for Socket Contact Engagement Test.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrandale, PA 15096-0001.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)



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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Classification of requirements. The requirements for the connectors are classified herein as follows:

| Requirement | Paragraph |
|---|---|
| Qualification | 3.3 |
| Materials | 3.4 |
| Design and construction | 3.5 |
| Performance | 3.6 |

3.3 Qualification. Connectors furnished under this specification shall be products which are qualified for listing on the applicable Qualified Product List (QPL) at the time set for the opening of bids (see 4.5 and 6.3).

3.4 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification and the applicable specification sheet. Acceptance or approval of any constituent material shall not be construed as a guaranty of acceptance of the finished products.

3.4.1 Dissimilar metals. Dissimilar metals are defined in MIL-HDBK-454. Unless suitably protected against galvanic corrosion by a compatible interposing metal, dissimilar metals shall not be used in intimate contact with each other.

3.4.2 Metal parts. Unless otherwise specified, metals parts, except spring members and hermetically sealed assemblies shall be a suitable copper alloy such as brass, in accordance with ASTM B16, ASTM B36, ASTM B121, and ASTM B124.

3.4.2.1 Spring members. Signal carrying spring members shall be manufactured of beryllium copper in accordance with ASTM B194, ASTM B196, and ASTM B197. Non-signal carrying spring member shall be manufactured of stainless steel in accordance with ASTM A484, or beryllium copper in accordance with ASTM B194, ASTM B196, and ASTM B197.

3.4.2.2 Hermetically sealed assemblies. Hermetically sealed assemblies are to be manufactured of steel in accordance with ASTM A108, or stainless steel in accordance with ASTM A484. Hermetically sealed assemblies may also be manufactured in a suitable sleeve before soldering into a copper alloy shell. Pin contacts may be of a ferrous alloy.

3.4.3 Plating.

3.4.3.1 Center contacts. Center contacts shall be .000050 (1.27μm) inch minimum gold in accordance with ASTM B488, type 3, grade C, class 1.27, over a suitable underplate except silver which shall not be used.

3.4.3.2 Shield clamp. Metal shield clamping mechanisms shall be .0001 inch minimum silver in accordance with QQ-S-365, type II, grade A. A suitable underplate may be used.

3.4.3.3 Outer contact (shell). Outer contacts of all types, shall be .000050 (1.27μm) inch minimum gold in accordance with ASTM B488, type 3, grade C, class 1.27, over a suitable underplate, except silver which shall not be used.

3.4.4 Insulating materials. Insulating materials shall be suitable for the purpose intended.

3.4.5 Fungus-inert materials. Materials which are not nutrients for fungus shall be used as indicated in MIL-HDBK-454.

3.4.6 Solder. Soft solder shall conform to ANSI J-STD-004, ANSI J-STD-005, and ANSI J-STD-006.

3.4.7 Flux. Flux, used to facilitate silver soldering, shall conform to O-F-499.



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3.5 Design and construction.

3.5.1 General design. The connectors shall be design for use with small radio frequency (rf) coaxial cables or shielded or unshielded wire as shown on the specification sheet (see 3.1). Solder pot of contact shall accommodate conductor sizes AWG 18, and smaller. Outer contact spring members shall be attached to the body for uninterrupted shielding and continuity under vibration conditions. Clamping of the cable shield to the connector body shall be independent of the type of rubber utilized, but shall provide metal-to-metal pressure and continuity. Polarization positions, as shown on figure 1, with mating lugs on the receptacles and matching slots on the plug couplings, shall be provided on the connectors. The retaining nut of the cable clamp shall include a suitable chamfer in the cable entry hole to prevent damage to the cable when exposed to shock and vibration.

3.5.2 Mating dimensions. The connectors shall be designed with mating dimensions in accordance with MIL-STD-348. The plug shall be so designed as to give a .005 inch minimum compression to the interface gasket when mated.

3.5.3 Moisture sealing. Moisture sealing shall be accomplished over the cable jacket in the cable entry area of the connector and over the cable primary insulation (core). Sealing shall also be effected in the area where the plug body fits the receptacle.

3.5.4 Contacts. All center contacts shall be of the “captive contact” design to insure electrical continuity without depending upon cable stability.

3.5.4.1 Socket contacts. The entering end of the socket contacts shall be rounded or chamfered to facilitate entrance of the mating contact.

3.5.5 Screw threads. Unless otherwise specified, screw threads shall conform to FED-STD-H28.



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3.5.6 Maintenance. The connector shall have a minimum number of parts consistent with reliability. Where practicable, the design shall permit easy assembly, disassembly, location of trouble sources, and maintenance with tools and equipment normally available commercially by service maintenance personnel with a minimum of training. Plugs and receptacles (series II) shall be assembled to cables by military standard hand operated crimp tools.

3.5.7 Interchangeability. All connectors of the same PIN shall be functionally and dimensionally interchangeable.

3.6 Performance.

3.6.1 Insulation resistance for mated connectors. When measured in accordance with 4.7.2, the insulation resistance of mated connectors shall conform to the minimum values of table II.

TABLE II. Insulation resistance in megohms.

| Class | Contacts | At ambient | At 200°C | After conditioning |
|---|---|---|---|---|
| G | Center contact to outer contact | 5,000 | 2,000 | 200 |
| All | Outer contact to outer shell (triaxial connector) | 1,000 | 400 | 40 |
| H, Y | Center contact to outer contact (hermetic connector) | 5,000 | 1,000 | 200 |

3.6.2 Dielectric withstanding voltage. When tested in accordance with 4.7.3, mated and unmated connectors shall conform to the minimum values specified in table III with no evidence of breakdown. Test potential is between center contact and outer contact. For type 2 and type 6 connectors, an additional potential shall be applied between the outer contact and the outer shell. For type 3 connectors, an additional potential shall be applied between the cable shield and outer contact.

TABLE III. Test voltage ac, rms, 60 cycle.

| Altitude | Unmated | Mated | Outer insulator type 2 and 6 | Interrupted ground type-3 |
|---|---|---|---|---|
| Sea level | 1,500 | 1,500 | 1,500 | 1,500 |
| 100,000 | 125 | 750 | 125 | 750 |

3.6.3 Contact resistance. When measured as specified in 4.7.4, contact resistance shall not cause the millivolt drop to exceed the values specified in table IV, as applicable.

TABLE IV. Contact resistance.

| Class | Type | Circuit | Test current | mV at 25°C | mV at 200°C | mV after conditioning |
|---|---|---|---|---|---|---|
| G | 1, 2, 3, 4 | Inner | 1A | 1.6 | 3.1 | 2.6 |
| | 1, 2 | Outer 1/ | 1A | 1.6 | 3.1 | 2.6 |
| H | 1, 2 | Inner | 1A | 16 | 32 | 32 |
| | | Outer 1/ | 1A | 1.6 | 3.2 | 3.2 |
| Y | 5, 6 | Inner | 1A | 3.0 | 60 | 60 |
| | | Outer | 1A | 2 | 4 | 4 |

1/ Does not apply for connectors using nonshielded cable.
NOTE: Allowable millivolt drop is doubled in the case of two contacts in series.



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3.6.4 Low signal level contact resistance. When tested as specified in 4.7.5, the low signal level contact resistance of each mated contact pair shall not exceed the applicable values specified in table V.

TABLE V. Low signal level contact resistance.

| Circuit (class A) | Milliohms maximum | |
|---|---|---|
| | Initial | After conditioning |
| Inner contact | 9 | 11 |
| Outer contact | 5 | 6 |

3.6.5 Contact engagement and separation forces. When tested in accordance with 4.7.6, the force required to insert and withdraw the specified pin shall be initially 18 ounces maximum and 0.7 ounce minimum. After conditioning, the maximum engagement force shall be 22 ounces and minimum separation force shall be 0.6 ounce.

3.6.6 Resistance to test probe damage. The socket contacts shall meet the contact resistance requirements after testing in accordance with 4.7.7. The entrance to the socket contact shall be designed to reject the entrance to a NO-GO test pin which is 0.044 ± .001 inch maximum diameter.

3.6.7 Contact retention. In unmated but assembled connectors, the contacts shall withstand an axial load of 10 pounds when tested in accordance with 4.7.8. Maximum allowable axial displacement of the contact after this test shall not exceed 0.010 inch.

3.6 8 Durability. It shall be possible to mate and unmate the connectors without deleterious effect on the mechanical or electrical performance when tested in accordance with 4.7.9.

3.6.9 Cable retention force (except classes H and Y). When properly attached to the applicable cable, the connector shall withstand a pull force of 30 pounds or 90 percent, whichever is less, of the breaking strength of the cable when tested in accordance with 4.7.10.

3.6.10 Side load (except classes H and Y). There shall be no evidence of bending or breaking of any part of the connector when tested in accordance with 4.7.11.

3.6.11 Hermetic sealing (except class G). Connectors which are designed with glass-to-metal seals for hermetic sealing applications shall not leak in excess of $1x10^{-5}$ cc/second of helium at a pressure differential of one atmosphere when tested in accordance with 4.7.12.

3.6.12 Thermal shock. Connectors shall withstand the shock or repeated surface exposures to extremes of high and low temperatures without affecting the mechanical or electrical characteristics when tested in accordance with 4.7.13.

3.6.13 Shock. Connectors shall withstand a 50g shock load when tested in accordance with 4.7.14 without interruption of signal for more than 10 microseconds. In addition, they shall sustain the shock load without impairing mechanical or electrical performance.

3.6.14 Vibration. When tested in accordance with 4.7.15, connectors shall not crack, nor break, and there shall be no loosening of parts. Connectors shall be in full engagement during vibration and the coupling device shall not loosen as a result of vibration. Interruptions of electrical continuity shall not last longer than 10 microseconds.

3.615 High temperature. After testing in accordance with 4.7.16, mated connectors shall have an insulation resistance of 2,000 megohms. Contact resistance shall be in accordance with table IV.

3.5.16 Fuel immersion. When tested in accordance with 4.7.17, electrical or mechanical performance shall not be impaired.

3.5.17 Altitude humidity cycling. All connectors shall withstand the electrical integrity portion of the altitude humidity cycling test as specified in 4.7.18.2.

3.6.18 Ozone exposure. After ozone exposure, as specified in 4.7.19, rubber parts shall show no evidence of cracking or other damage that will affect performance adversely. A certified statement of the materials manufacturer may be submitted in lieu of performing the test.

3.6.19 Corrosion resistance. After passing the corrosion test specified in 4.7.20, the base metal of plated parts of the connectors shall not be exposed.



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3.6.20 Sulphide immersion. The finish of mated connectors shall not become tarnished when tested in accordance with 4.7.21.

3.6.21 Maintenance aging. When connectors are tested as specified in 4.7.22, they shall show no evidence of severe mechanical damage and the coupling device shall remain functional.

3.6.22 Shield clamp resistance (except classes H and Y). The maximum resistance shall be as specified (see 3.1) when tested in accordance with 4.7.23.

3.7 Marking. Connectors and associated fittings shall be permanently and legibly marked in accordance with the general marking requirements of MIL-STD-130 the military PIN (see 1.2.1) and the manufacturer's federal supply code (H4-1). The marking location is optional; when practicable, a location should be picked that will least likely be covered in cable assembly or installation.

3.8 Workmanship. The connectors, including all parts and accessories, shall be constructed and finished in a throughly workman like manner. Particular attention shall be paid to neatness and thoroughness of soldering, freedom from burrs, and sharp edges, and freedom from loose, spattered, or excess solder, metal chips and other foreign material.

3.9 Assembly instructions. Complete assembly instructions shall be furnished by the vendor with each connector procured under this specification (see 6.4).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.5).
b. Conformance inspection (see 4.6).

4.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment (i.e., industry standard, MIL-STD) shall be required.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "General Requirements" of MIL-STD-1344.

4.4 Preparation of samples. Connectors, except class Y, shall be wired with three feet of wire or cable as shown in Appendix I.

4.4.1 Crimp terminations. Contacts and shield terminations shall be crimped as specified (see 3.1). When applicable, MIL-C-22520 tools shall be used.

4.4.2 Solder terminations. Soldering shall be in accordance with J-STD-004, J-STD-005, J-STD-006, or MIL-HDBK-454.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production. Application for qualification tests shall be made in accordance with Provisions Governing Qualification (SD-6) (see 6.3).

4.5.1 Sample size. Unless otherwise directed by the qualifying activity (see 6.3), the test samples shall consist of 12 connectors of one polarity which shall be submitted in equal quantities of compatible mating types. The samples shall be separate into the following test groups.

a. Eight sets of mated connectors, with associated cable or wire attached, shall be divided into test groups A and B of four sets each.
b. Four unmated connectors shall compose test group C. Cable types shall be wired in accordance with 4.5.2.
c. Twelve sets of mated contacts shall be divided evenly between groups D and E.

4.5.1.1 Non-removable contacts. Contacts for connectors that normally have nonremovable contacts shall be selected from contact lots that have been finished in the same manner as those in the connector. The connector shells and insulators shall not be included in the finishing process.

4.5.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in table VI, in the order shown.

4.5.3 Failures. One or more failures in any of the applicable inspection shall be cause for refusal to grant qualification.



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TABLE VI. Qualification inspection.

| Inspection | Requirement paragraph | Test method paragraph | Test groups | | | | |
|---|---|---|---|---|---|---|---|
| | | | A | B | C | D | E |
| Visual and mechanical inspection 1/ | 3.1, 3.4, 3.5, 3.7, 3.8 | 4.7.1 | 1 | 1 | 1 | 1 | 1 |
| Insulation resistance | 3.6.1 | 4.7.2 | 4 | 4 | 4 | | |
| Dielectric withstanding voltage | 3.6.2 | 4.7.3 | 4 | 4 | | | |
| Contact resistance | 3.6.3 | 4.7.4 | | 4 | | 6 | 6 |
| Low signal level contact resistance | 3.6.4 | 4.7.5 | | 4 | | 6 | 6 |
| Contact engagement and separation forces | 3.6.5 | 4.7.6 | | | | 6 | |
| Resistance to test probe damage | 3.6.6 | 4.7.7 | | | 4 | | |
| Contact retention | 3.6.7 | 4.7.8 | | | 4 | | |
| Durability | 3.6.8 | 4.7.9 | | | 4 | | 6 |
| Cable retention force (except classes H and Y) | 3.6.9 | 4.4.10 | 4 | | | | |
| Side load (except classes H and Y) | 3.6.10 | 4.7.11 | | | 4 | | |
| Hermetic sealing (except class G) | 3.6.11 | 4.7.12 | | 4 | | | |
| Thermal shock | 3.6.12 | 4.7.13 | | 4 | | | |
| Shock | 3.6.13 | 4.7.14 | | 4 | | | |
| Vibration | 3.6.14 | 4.7.15 | | 4 | | | |
| High temperature | 3.6.15 | 4.7.16 | | 4 | | 6 | |
| Fuel immersion | 3.6.16 | 4.7.17 | 4 | | | 6 | |
| Altitude humidity cycling | 3.6.17 | 4.7.18 | 4 | | | | |
| Ozone exposure | 3.6.18 | 4.7.19 | | | 4 | | |
| Corrosion resistance | 3.6.19 | 4.7.20 | | 4 | | 6 | |
| Sulphide immersion | 3.6.20 | 4.7.21 | 4 | | 4 | | |
| Maintenance aging | 3.6.21 | 4.7.22 | 4 | 4 | | | |
| Shield clamp resistance (except classes H and Y) | 3.6.22 | 4.7.23 | 2 | 2 | | | |

1/ Marking will be considered defective only if it is illegible at the completion of the required tests.

4.5.4 Retention of qualification. To retention qualification, the contractor shall verify in coordination with the qualifying activity the capability of manufacturing products which meet the performance requirements of this specification. Refer to the qualifying activity for the guidelines necessary to retain qualification to this particular specification. The contractor shall immediately notify qualifying activity any time that the inspection data indicates failure of the qualified product to meet the performance requirements of this specification.

4.5.5 Qualification of additional polarities. Qualification may be granted for all polarities of one series, class, and type of connector, provided connector samples of each of the four bayonet configurations have passed the tests in table VI in a different series, class, or type.

4.5.6 Qualification of additional cable sizes (class G and H). Qualification may be granted for additional cable sizes provided those cable sizes have passed the tests in table VI in a different type configuration. For additional cable sizes not previously tested, four mated connector samples shall be subjected to the inspections specified in table VII.

TABLE VII. Qualification of additional cable sizes.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Visual and mechanical inspection 1/ | 3.1, 3.4, 3.5, 3.7, 3.8 | 4.7.1 |
| Insulation resistance | 3.6.1 | 4.7.2 |
| Dielectric withstanding voltage | 3.6.2 | 4.7.3 |
| Cable retention force (except classes H and Y) | 3.6.9 | 4.7.10 |
| Side load (except classes H and Y) | 3.6.10 | 4.7.11 |
| Corrosion resistance | 3.6.19 | 4.7.20 |
| Shielded clamp resistance (except classes Hand Y) | 3.6.22 | 4.7.23 |

1/ Marking will be considered defective only if it is illegible at the completion of the required tests.



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4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all connectors of the same PIN produced under essentially the same conditions, and offered for inspection at one time.

4.6.2 Group A inspection. Group A inspection shall consist of the inspections specified in table VIII in the order shown.

TABLE VIII. Group A inspection.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Visual and mechanical inspection 1/ | 3.1, 3.4, 3.5, 3.7, 3.8 | 4.7.1 |
| Dielectric withstanding voltage 2/ | 3.6.2 | 4.7.3 |
| Contact engagement and separation force | 3.6.5 | 4.7.6 |
| Hermetic seal (except class G) | 3.6.11 | 4.7.12 |

1/ Marking will be considered defective only if it is illegible at the completion of the required inspections.
2/ Does not apply to connectors with all contacts and insulation unassembled.

4.6.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IX, in the order shown. Sample units shall be selected from inspection lots that have passed group A inspection.

TABLE IX. Group B inspection.

| Inspection | Requirement paragraph | Test method paragraph | Mated group A | Not mated group B | Contacts group C |
|---|---|---|---|---|---|
| Visual and mechanical inspection 1/ | 3.1, 3.4, 3.5, 3.7, 3.8 | 4.7.1 | X | X | X |
| Maintenance aging | 3.6.21 | 4.7.22 | X | | |
| Insulation resistance | 3.6.1 | 4.7.2 | X | | |
| Dielectric withstanding voltage | 3.6.2 | 4.7.3 | X | X | |
| Contact resistance | 3.6.3 | 4.7.4 | X | | X |
| Side load (except classes H and Y) | 3.6.10 | 4.7.11 | X | | |
| Fuel immersion | 3.6.16 | 4.7.17 | X | | |
| Corrosion resistance | 3.6.19 | 4.7.20 | | X | X |

1/ Marking will be considered defective only if it is illegible at the completion of the required tests.

4.6.3.1 Sampling plan. Sample units of the same PIN, representative of production at the time of selection shall be selected at six month intervals for 1 year. Upon passing this inspection, contractor may select sample units every 12 months. If the second level of sampling is passed two successive times, the contractor may select sample units every 24 months. In the event of a failure, sampling shall revert to the 6-month interval.

4.6.3.2 Failures. If one or more units fail to pass group B inspection, the sample shall be considered to have failed.

4.6.3.3 If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.3.4 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.



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4.6.3.5 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subjected to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Group B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7 Methods of inspection.

4.7.1 Visual and mechanical examination. Connectors shall be inspected to verify that the materials, screw threads, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.7.2 Insulation resistance. The connector shall be measured for insulation resistance between the center conductor and connector body, or inner and outer conductors of wired and mated assemblies, at 500 volts dc in accordance with method 3003 of MIL-STD-1344 (see 3.6.1).

4.7.3 Dielectric withstanding voltage. A dielectric withstanding voltage test for both mated and unmated connectors shall be conducted in accordance with method 3001 of MIL-STD-1344. The test voltage shall be in accordance with table III (see 3.6.2).

4.7.4 Contact resistance. Electrical resistance of mated connector inner ands outer contracts shall be determined separately in accordance with method 3002 of MIL-STD-1344. Test current shall be in accordance with table IV (see 3.6.3).

4.7.4.1 Inner contacts. A test current as required by table IV shall be applied to the ends of the wire attached to the mated contacts. One test probe shall be placed on each end of the mated contact pair and the millivolt drop shall be measured by means of a suitable instrument.

4.7.4.2 Outer contacts. Test as above, except that the current is applied to the coaxial shield of the attached cables, and the voltage test probes placed on each end of the mated connector pair.

4.7.5 Low signal level contact resistance. When tested in accordance with method 3002 of MIL-STD-1344, the low signal level contact resistance of each mated pair shall not exceed the applicable values specified in table V (see 3.6.4).

4.7.6 Contact engagement and separation forces. When tested in accordance with method 2014 of MIL-STD-1344, the force levels shall comply with 3.6.5. A 0.0390 +.0000 -.0001 inch diameter pin shall be inserted and removed from each socket contact. The engagement forces shall be measured during insertion. A 0.0370 +.0001, -.0000 inch diameter pin shall be inserted and removed from each contact and the separation force shall be measured during removal. Except as noted herein, the pins shall conform to SAE-AS31971. The insertion depth shall be 0.125 inch minimum.

4.7.7 Resistance to test probe damage (see 3.6.6). Socket contacts shall be tested in accordance with method 2006 of MIL-STD-1344. The following details and exceptions shall apply:

a. The contact shall be locked into a connector.
b. The test probe shall have a diameter of 0.038 ±0.001 inch.
c. While inserted, a bending moment of ½ inch-pound ±10 percent shall be applied to the probe.
d. After the test, the contacts shall meet the requirements of 3.5.5.
e. The NO-GO test pin (0.044 ± .001 inch diameter) shall be applied with a force of 5 ±1 pounds.

4.7.8 Contact retention (see 3.6.7). An unmated connector assembled with its proper cable shall be tested in accordance with method 2007 of MIL-STD-1344. The following details shall apply:

a. The axial load shall be 10 pounds.
b. The load shall be applied from the mating end.
c. The axial displacement shall be in accordance with 3.6.7.

4.7.9 Durability (see 3.6.8). Parts of mated connectors shall be tested in accordance with method 2016 of MIL-STD-1344. The connectors shall be mated and unmated 500 times.



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4.7.10 Cable retention force (except classes H and Y). The connector shall be properly wired to a 3-foot length of cable and connected in a test circuit to indicate any discontinuity. It shall then be subjected to an axial pull force which will tend to pull the attached cable from the connector. The cable shall not separate from the connector nor indicate discontinuity between the shield and the shell or the center conductor and the center contact (see 3.6.9).

4.7.11 Side load (except classes H and Y) (see 3.6.10).

4.7.11.1 Cable plugs. Cable plugs and right-angle cable plugs shall be mated with a fixed, mounted receptacle and shall withstand a side-load force of 30 pounds or 90 percent of breaking strength of cable applied at the cable end and at right angles to the major axis of the mounted jack (see figures 2a and 2b).

4.7.11.2 Cable receptacles. A side-load test shall not be required for cable receptacles.

4.7.11.3 Panel mounting receptacles. Panel mounting receptacles shall be mounted on a panel and shall withstand a side-load force of 30 pounds or 90 percent of breaking strength of cable applied at the cable end and at right angles to the major axis (see figure 2c).

4.7.11.4 Adapters. Adapters shall be secured to a fixed, mounted receptacle or clamped in a vise and shall withstand a side-load force of 25 pounds applied at the other end and at right angles to the major axis (see figure 2d).

4.7.12 Hermetic sealing (except class G) (see 3.6.11). Connectors shall be in accordance with method 1008 of MIL-STD-1344. The following detail shall apply:

The pressure differential shall be one atmosphere.

4.7.13 Thermal shock. A pair of mated and wired connectors shall be tested in accordance with method 1003.1, test condition A, of MIL-STD-1344 (see 3.6.12).

4.7.14 Shock (see 3.6.13). A pair of wired and mated connectors shall be tested in accordance with method 2004, test condition A, of MIL-STD-1344. The following details and exceptions shall apply:

a. Connectors shall be mounted on the shock device or carriage as shown on figure 3 of the vibration test.
b. The connector shall be fully wired and the cable clamped to points that move with the connector. A minimum of 8 inches of cable shall be unsupported behind the rear of each connector.
c. Inner and outer contacts shall be wired in series and a test current of 100 milliamperes flowing during the test. The open circuit voltage shall be 6 volts. A check shall be made for interruption of the test current.
d. After testing, the connectors shall have remained mated and mechanically functional.

4.7.15 Vibration. The connector assembly shall be mounted as shown on figure 3 and vibrated in accordance with method 2005, condition IV, of MIL-STD-1344. In addition, vibration shall be conducted at a low temperature ambient of –55°C and a high ambient of 200°C. Inner and outer contacts shall be wired in series with at least 100 milliamperes of current flowing. The open circuit test voltage shall be 6 volts. A suitable instrument shall be used to monitor current flow and to indicate any discontinuity of contact or interruption of current flow. Duration of vibration at extremes of temperature shall be 25 percent of the duration specified for the standard temperature condition. The connectors shall remain in full engagement during vibration. The coupling device shall not loosen and there shall be no cracking or breaking of parts (see 3.6.14).

4.7.15.1 Connector mounting. The vibration mounting shall be in accordance with figure 3 except as specified herein. The receptacle shall be mounted on an adapter plate, using the normal mounting provisions and suitable hardware. The connector mounting bracket may be of another design, provided the connectors received the required magnitude of vibration. The associated plug or adapter shall be mated with the receptacle, as used in service. Safety wires or supplementary locking devices shall not be used.

4.7.16 High temperature. Mated connectors shall be subjected to 200°C for 1 hour. At the end of this time and while at the elevated temperature the connectors shall be subjected to the insulation resistance test 4.7.2 and the contact resistance test 4.7.4 (see 3.6.15).

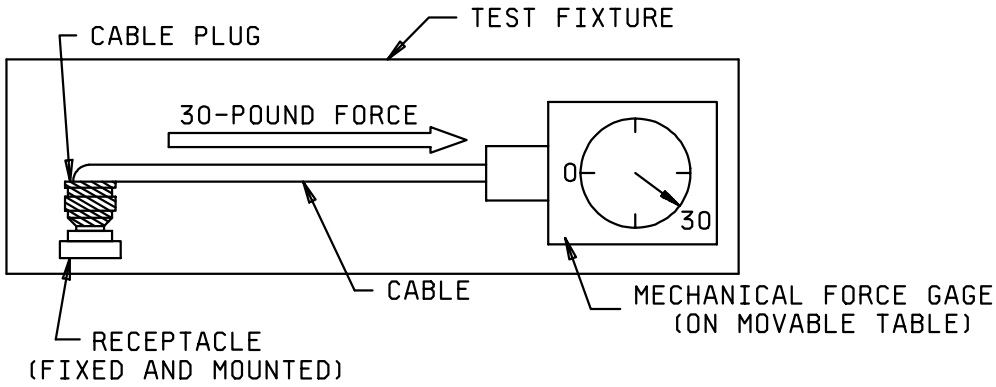
4.7.17 Fuel immersion. Mated and cabled connectors shall be immersed in type II hydrocarbon fuel in accordance with ASTM D471 at +71°C for 7 days. Following this test, the connectors shall unmate and mate satisfactorily with their corresponding plug or receptacle and pass the insulation resistance, dielectric withstanding voltage and contact resistance tests (see 3.6.16).

4.7.17.1 Fluids. Separate samples of each of the nonmetallic parts of the connectors shall be immersed in each of the fluids listed in table X for the specified time and temperature. At the end of this period, all parts shall be rinsed, air blasted, and dried at 40°C for 24 hours. A visual inspection shall be made to insure that there is no cracking, peeling, splitting, blistering, or swelling.



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(a) CABLE PLUGS




(b) RIGHT ANGLE CABLE PLUGS

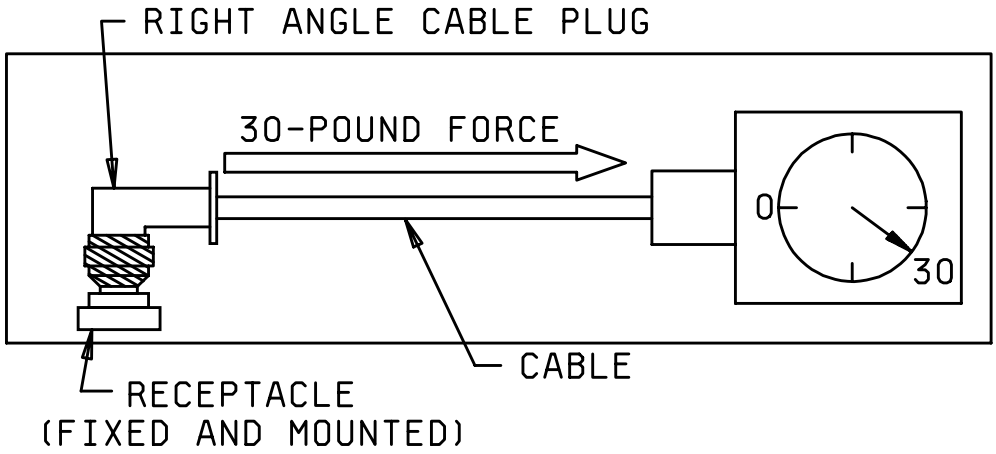
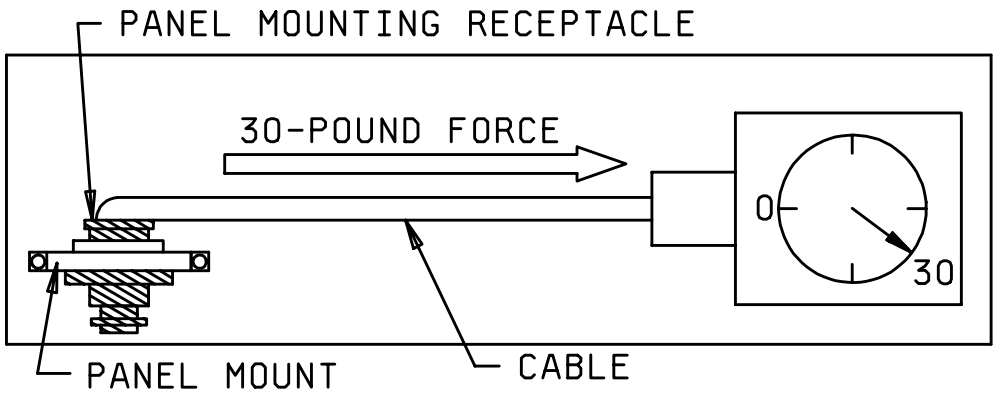



FIGURE 2. Side and test diagram.



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(c) PANEL MOUNTING RECEPTACLES




(d) ADAPTERS

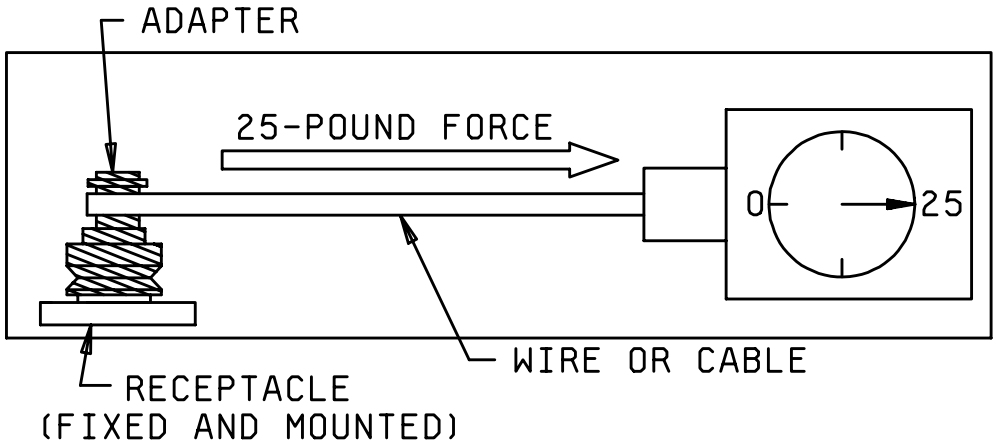
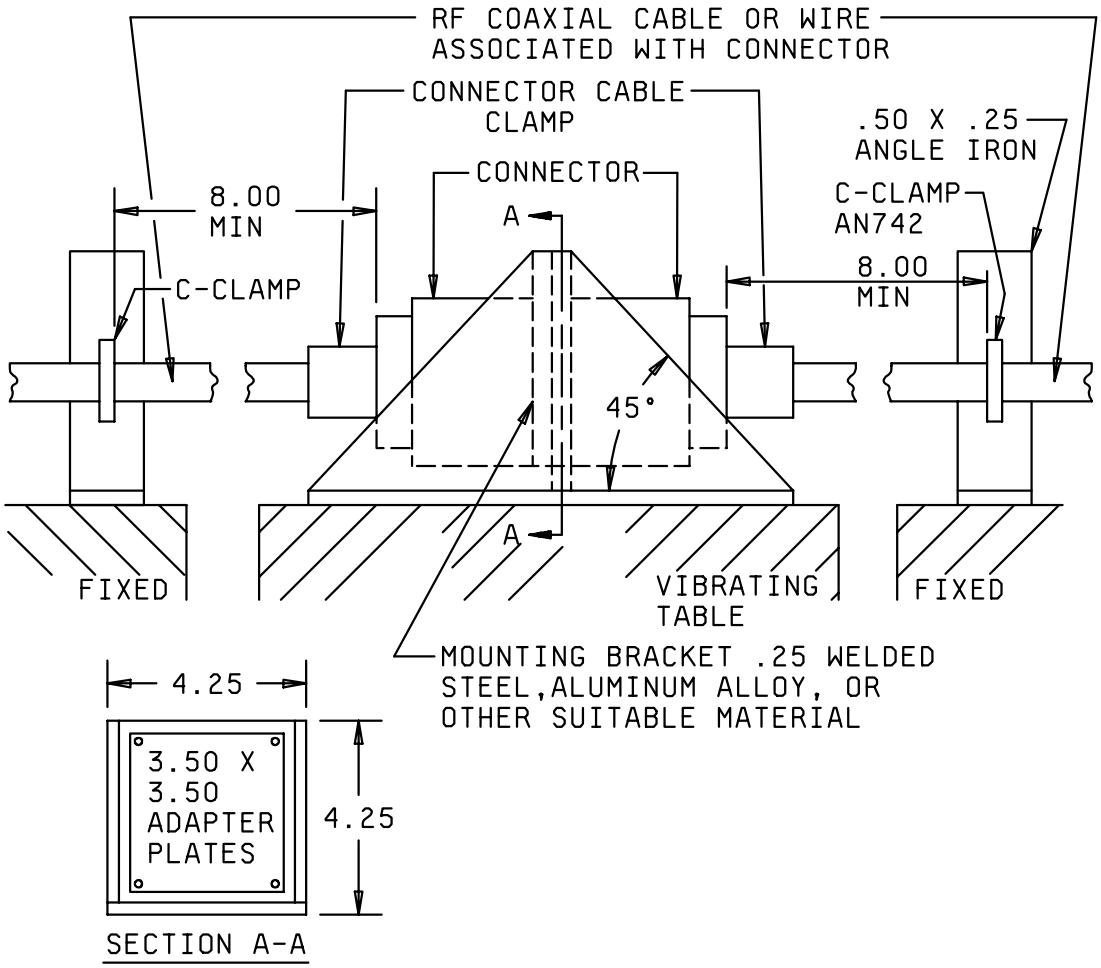



FIGURE 2. Side and test diagram - Continued.



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| Inches | mm |
|---|---|
| .25 | 6.3 |
| .50 | 12.7 |
| 3.50 | 88.9 |
| 4.25 | 108.0 |
| 8.00 | 203.2 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Vibration testing arrangement.



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TABLE X. Fuels.

| Fluids | Specification | Temp °F | Immersion time (hours) |
|---|---|---|---|
| Aviation gasoline | ASTM D910 | 160 | 168 |
| Aviation fuel (grade JP-4) | MIL-DTL-5624 | 160 | 168 |
| Oil, lubricating, jet engine (grade 1010, petroleum base) | MIL-PRF-6081 | 160 | 20 |
| Lubricating oil, aircraft | MIL-L-7808 | 160 | 20 |
| Salt water (95 percent water, 5 percent chloride) | | 160 | 20 |
| Diesel, fuel oil | A-A-52557 | 160 | 168 |
| Hydrocarbon fluid | ASTM D471 | 160 | 168 |
| Aviation hydraulic fluid, petroleum base | MIL-H-5606 | 160 | 20 |

4.7.18 Altitude humidity cycling (see 3.6.17). Connectors shall be mounted on a sealed test vessel and subjected to alternate vacuum-pressure environment in the presence of test fluid and water vapor at both high and low temperatures to determine leak integrity and the effects on electrical characteristics. Figure 4 shows the recommended test vessel.

4.7.18.1 Leak integrity.

a. Hermetically sealed and pressurized connectors shall be installed approximately as shown on the right side of the pressure-tight vessel as shown on figure 4.
b. The test box shall be filled with test fluid, in accordance with type III of ASTM D471, to a level of a least 4 inches above the connectors.
c. With the fluid in the test box at room temperature, 6 pounds per square inch pressure ($lb/in^2$) shall be applied to the box and held for 1 hour while checking for leaks.
d. With the test box at room temperature, pressure shall be slowly increased to 20 $lb/in^2$ and then relieved. This cycle shall be repeated for a total of 10 times. On the tenth cycle, the pressure shall be held at 20 $lb/in^2$. A check shall be made for signs of leakage and pressure shall be relieved.
e. The temperature of the test box shall be decreased to –54°C and pressure of 6 $lb/in^2$ shall be applied and held for 1 hour while checking for leaks. The temperature shall be held at –54°C and the pressure slowly increased to 20 $lb/in^2$ and then relieved. This cycle shall be repeated for a total of 10 times. On the tenth cycle, the pressure shall be held at 20 $lb/in^2$. A check shall be made for leakage. Pressure shall be relieved.
f. The temperature of the test box shall be allowed to return to room temperature and, with pressure relieved, the wire in the connector shall be pulled from the inside of the test box with a force of 5 ±1/2 pounds. (For this test, a mechanical device may be rigged to pull the cable or contact from the inside without removing the cover.). With the test box still at room temperature, the box shall be pressurized at 6 $lb/in^2$ pressure and a check made for leaks. The pressure shall be increased to 20 $lb/in^2$ and a check made for leaks. Upon completion of these tests, the pressure shall be relieved.

4.7.18.2 Electrical integrity.

a. The connectors shall be removed from the test box, examined for damage, and remounted approximately as shown on the left side of figure 4 so that the connector receptacle is toward the inside of the test vessel. Both ends of the electrical wire shall terminate outside the test box. Water, containing 5 percent salt solution, shall be poured inside the box to a fluid level at least 1 inch below the connectors.

b. A vacuum of 27 inches of mercury shall be applied to the test box for 5 minutes and then relieved. The box shall be placed in an ambient temperature of 71°C for 1 hour and then reduced to -1°C for 1 hour. Insulation resistance shall be checked at 500 volts dc, using the ends of the outside lead wire and the body of the connector as terminals. The insulation resistance shall not read less than 200 megohms. This cycle shall be repeated for a total of 10 minutes.



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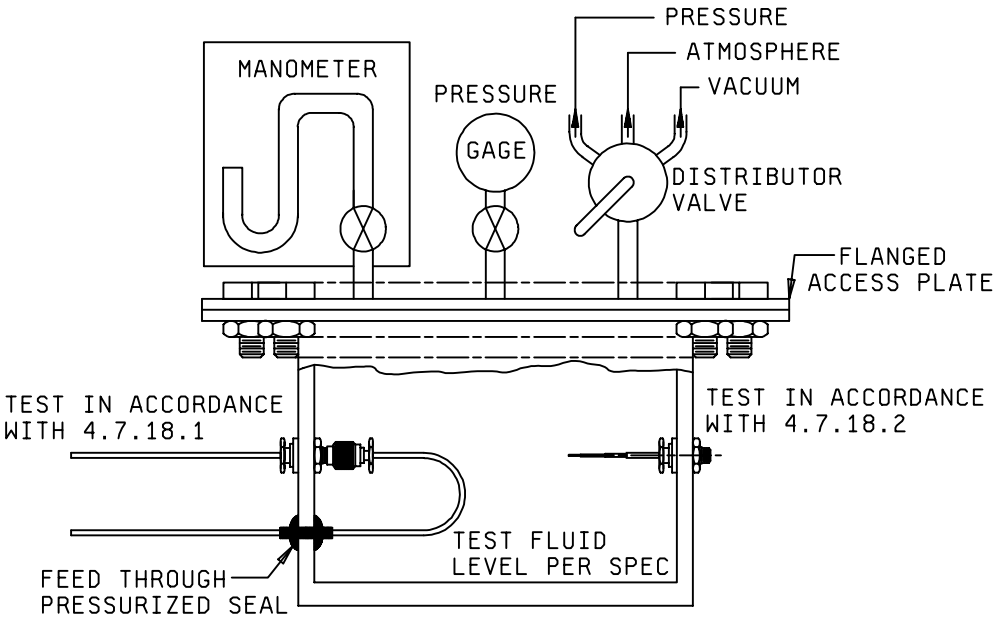



FIGURE 4. Sealed test vessel.



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4.7.19 Ozone expose. Connectors shall be tested for ozone exposure in accordance with method 1007 of MIL-STD-1344. Rubber portions of the connectors shall be subjected to ozone exposure having a concentration of 0.010 to 0.015 percent by volume for 2 hours at room temperature. After testing, the connectors shall be visually checked for evidence of cracking or other damage that will affect performance adversely (see 3.6.18).

4.7.20 Corrosion resistance. Mated and wired connectors and contacts shall be subjected to 5 percent salt spray in accordance with method 1001.1, condition B of MIL-STD-1344. Mated connectors shall pass the insulation resistance test (4.7.2) and outer contact resistance test (4.7.4). It shall be possible to mate and unmate the connectors as in service and there shall be no corrosion (3.6.19).

4.7.21 Sulphide immersion. Mated and cabled connectors shall be immersed for 1 minute in the following solution, prepared as specified, after which they shall be double rinsed in tap water, rinsed in alcohol, air dried, and examined for tarnish:

a. Dissolve 1 pound of sodium sulphite in 800 $cm^3$ of distilled water.
b. Dilute to 1000 $cm^3$.
c. Add 250 grains of powdered sulphur.
d. Allow solution to stand 16 hours.
e. Dilute with distilled water to a specific gravity of 1.14. This solution shall be replaced once each week.

4.7.22 Maintenance aging. The connectors shall be mated and unmated 10 times to provide accelerated maintenance aging of the contact, contact locking mechanism, and sealing provisions (see 3.6.21).

4.7.23 Shield clamp resistance (except classes H and Y) (see 3.6.22). The shield clamp (cable braid to connector) resistance measurement shall be conducted with the apparatus shown on figure 5 and under the steps indicated below:

a. Accurately determine the resistance per inch of several feet of the coaxial cable shield (braid to conductor) and record this value as $R_b$/inch.
b. Assemble 5-inch lengths of cable to the test sample connectors and accurately determine the shield effective length in inches from where the shield enters the braid clamp to where it enters the power supply connection, as shown on figure 6 and record this value as Eb.
c. Then the effective cable shield (braid) resistance shall be ($R_b$/inch) (Eb) = $R_b$. (Eb is the effective braid length) ($R_b$ is the effective braid resistance).
d. Determine the total shield circuit resistance from the body of the connector to the power supply cable braid connection and record as $R_T$.
e. Then the resistance of the shield clamp connection will be $R_T - R_b = R_c$. $R_c$ is the shield clamp resistance in milliohms.

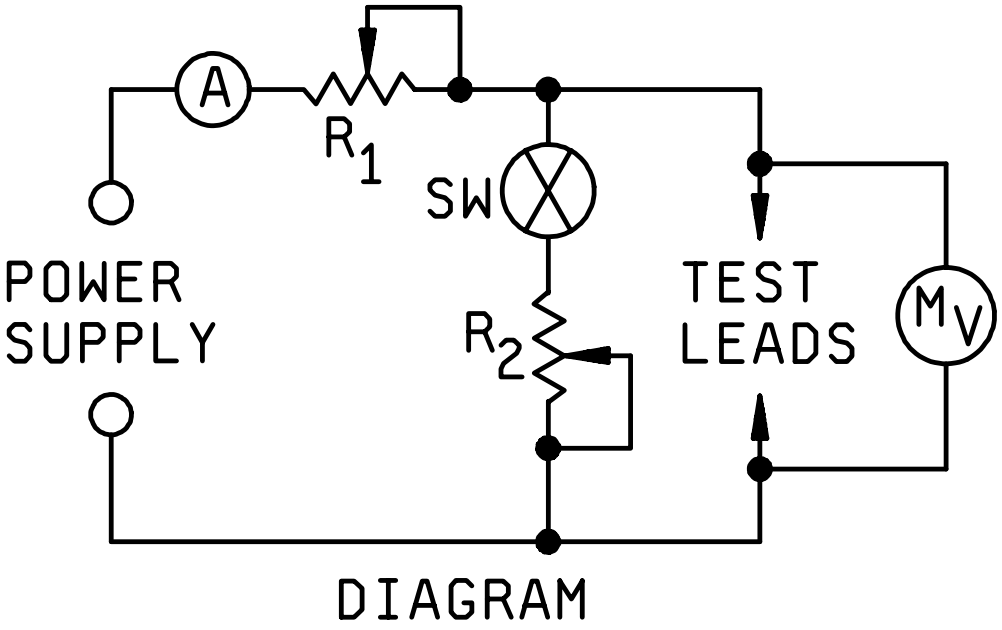
NOTE: The millivolt values can be used as milliohm values, if the current has been kept to one ampere as required.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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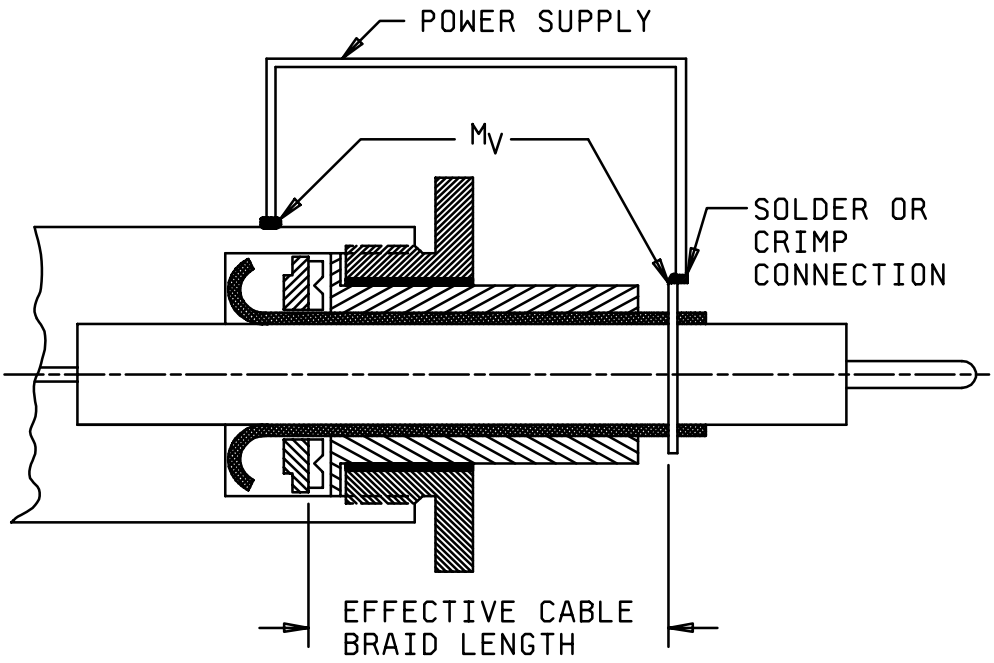

| | | | |
|---|---|---|---|
| A Ammeter | 0-1 Amp | R2 Variable resistor | 0-100 mil ohms |
| R1 Variable resistor | 0-15 Ohms | Mv Millivoltmeter | 0-100 mV |
| SW Switch | Less than .01 ohms Contact resistance | Power supply | 10 volts DC at 1 amp. |

a. With test leads open, close switch SW.
b. Adjust $R_2$ for a millivoltmeter reading of 50 mV.
c. Connect test leads across the test sample and see that the mV drops significantly before opening switch SW required in (d). A drop in mV will indicate satisfactory dry circuit continuity.
d. Open SW and then adjust $R_1$ for a circuit current A of one ampere.
e. Measure the millivolt drop mV across the test sample and call this ET.

FIGURE 5. Resistance measurement (voltage drop).



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NOTE: TYPICAL SHIELD CLAMP SHOWN (SPECIFIC DESIGN NOT REQUIRED).
ILLUSTRATES METAL TO METAL CLAMPING OF THE SHIELD (REQUIRED).

FIGURE 6. Set-up for resistance measurement of cable shield clamp connection.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The miniature coaxial connectors covered by this specification are intended primarily for use in capacitance-type fuel-quantity-measurement systems in all types of aircraft and missiles at the lower frequencies. These connectors are not intended to be used for RF applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL No 25516 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the qualified products lists is the Defense Supply Center Columbus (DSCC-VQP), 3990 East Broad Street, Columbus, Ohio 43216-5000.

6.4 Assembly instructions. Assembly instructions should include:

a. Cable preparation-stripping dimensions and tolerances.
b. List and description of crimping tools if required.
c. Pictorial presentation of sub-assemblies and loose piece parts.
d. Sufficient pertinent dimensions for verification of correct parts; as a minimum the cable entry for conductor, dielectric braid, and jacket shall be specified.
e. Recommended cable clamp tightening torque (if applicable).

6.5 Specific requirements. If a specific requirement specified in the general specification is not required for an item; it should be so indicated on the specification sheet.

6.6 Definitions.

Isolated ground – where system ground is separated from real ground.
Interrupted ground – where ground path is broken at one or both ends of cable.

6.7 Subject term (key word) listing.

Contact
Finish
Material
Spring members
Qualification

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy – EC
Air Force - 11
DLA- -CC

Preparing activity:
DLA - CC

(Project 5935-4214)

Review activities:
Navy - AS
Air Force - 99



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APPENDIX A

ASSOCIATED CABLE DIMENSIONS

| MIL-C-17 PIN M17/ REF 1/ | Cable designation dash digits | Center conductor solid or str. max. AWG size | Core O.D. min / max (inch) | Shielded | Jacket O.D. min / max (inch) |
|---|---|---|---|---|---|
| 84-RG223 | -01 | 20 | .109 / .119 | Shielded | .198 / .212 |
| 93-RG178 | -02 | 20 | .035 / .045 | Shielded | .070 / .085 |
| | -03 | 19 | .065 / .075 | Shielded | .110 / .120 |
| | -04 | 19 | 082 / .092 | Shielded | .120 / .130 |
| | -05 | 19 | .085 / .095 | Shielded | .130 / .140 |
| | -06 | 20 | .075 / .095 | Nonshielded | |
| | -07 | 19 | .090 / .100 | Shielded | .136 / .150 |
| | -08 | 20 | .044 / .054 | Shielded | .074 / .088 |
| | -09 | 20 | .052 / .062 | Shielded | .081 / .096 |
| | -10 | 20 | .056 / .060 | Nonshielded | |
| | -11 | 20 | .125 / .135 | Nonshielded | |
| | -12 | 19 | .090 / .100 | Nonshielded | |
| 119-RG174, 94-RG179, 113-RG316 | -13 | 20 | .050 / .060 | Shielded | .100 / .110 |
| 94-RG179 | -14 | 20 | .060 / .066 | Shielded | .095 / .105 |
| | -15 | 18 | .110 / .120 | Nonshielded | |



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APPENDIX A

ASSOCIATED CABLE DIMENSIONS

| MIL-C-17 PIN M17/ REF 1/ | Cable designation dash digits | Center conductor solid or str. max. AWG size | Core O.D. min/max (inch) | Shielded | Jacket O.D. min/max (inch) |
|---|---|---|---|---|---|
| 54-RG122 | -16 | 20 | .105/.115 | Shielded | .147/.165 |
| | -17 | 20 | .090/.100 | Nonshielded | |
| 95-RG180 | -18 | 20 | .100/.110 | Shielded | .145/.155 |
| 28-RG58, 111-RG303, 60-RG142 | -19 | 18 | .110/.121 | Shielded | .188/.202 |
| | -20 | 18 | .066/.070 | Nonshielded | |
| | -21 | 18 | .080/.090 | Nonshielded | |
| | -22 | 19 | .068/.070 | Shielded | .133/.147 |
| | -23 | 20 | .123/.133 | Shielded | .160/.170 |
| | -24 | 18 | .118/.128 | Shielded | .158/.172 |
| 95-RG180 | -25 | 20 | .105/.115 | Shielded | .130/.150 |



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## STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

### INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-DTL-25516E | 2. DOCUMENT DATE<br>2000/08/22 |
|---|---|---|

**3. DOCUMENT TITLE**
**Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification For**

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include zip code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) DSN *(if applicable)* | 7. DATE SUBMITTED *(YYYYMMDD)* |

**8. PREPARING ACTIVITY**

| a. NAME<br>**Defense Logistics Agency**<br>**Defense Supply Center, Columbus** | b. TELEPHONE *(Include Area Code)*<br>(1) Commercial **614-692-0538**<br>(2) DSN **850-0538** |
|---|---|
| c. ADDRESS *(Include Zip Code)*<br>**DSCC-VAI**<br>**P.O. Box 3990**<br>**Columbus, Ohio 43216-5000** | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>**Defense Standardization Program Office (DLSC-LM)**<br>**8725 John J. Kingman Road,Suite 2533**<br>**Fort Belvoir, Virginia 22060-6621**<br>**Telephone (703) 767-6888 DSN 427-6888** |

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